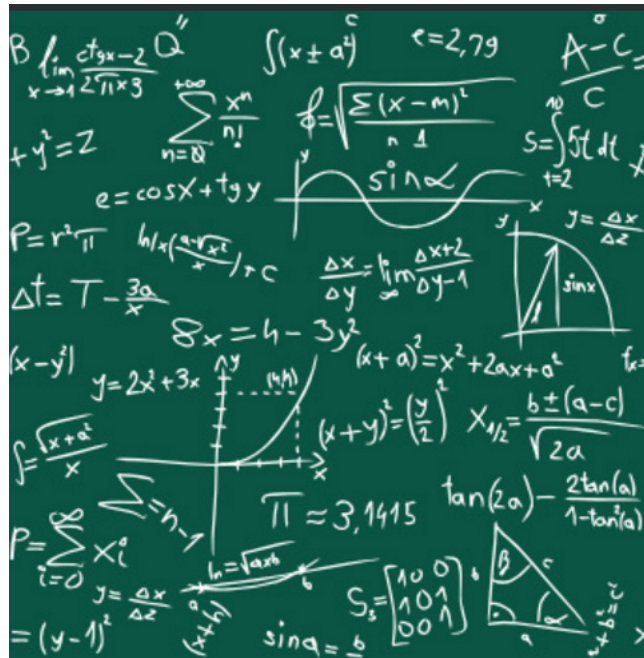


Functions



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 The concept of function

The existence of a **variable quantity** implies the existence of a **relation** between **two quantities**, that is, the dependence of one quantity on another.

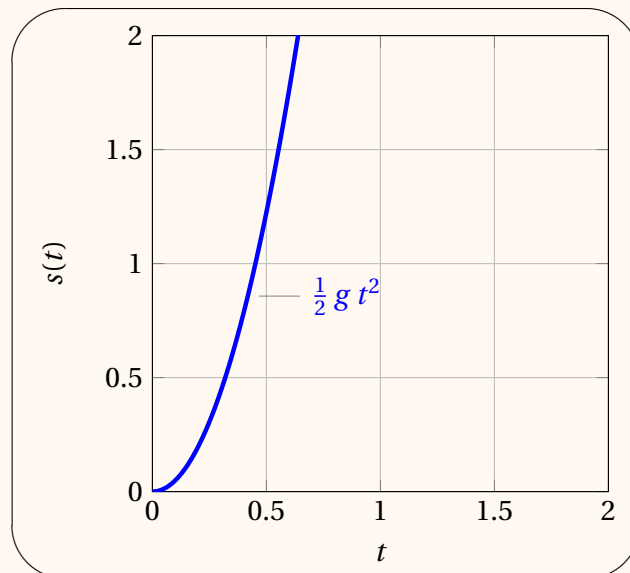
- In each case, this relation follows a certain **law** or **formula**.

Example 1: Laws/formulas $\rightarrow$ functions

If a heavy object is dropped from a certain height, the distance traveled by the object varies with time t according to the formula:

$$s(t) = \frac{1}{2} g t^2 \quad \text{with } t \geq 0$$

where $g \approx 9,8$ is the gravitational acceleration constant.



Hence, the time t is associated with the distance traveled $s(t)$: $t \mapsto s(t)$.

Example 2: Laws/formulas $\rightarrow$ functions

If a unit capital is invested for one year at the annual rate i , with interest paid monthly at a rate $\frac{i}{12}$, we have:

- after the first month, a capital equal to:

$$1 + \frac{i}{12} \cdot 1 = 1 + \underbrace{\frac{i}{12}}_{\alpha}$$

- after the second month, a capital equal to:

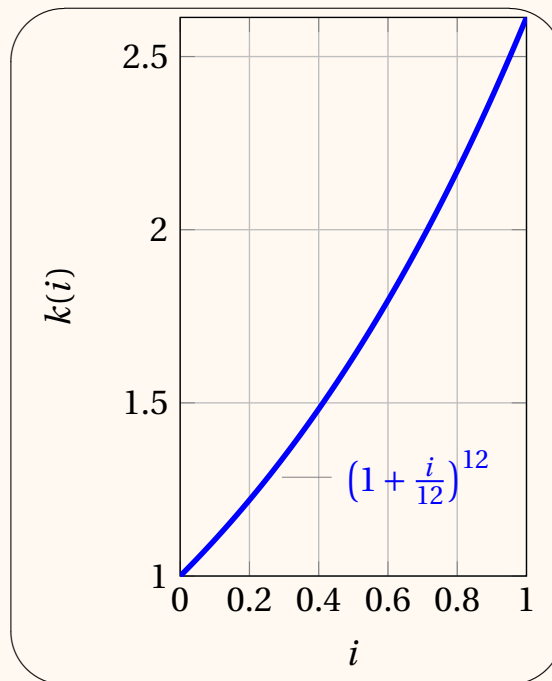
$$\underbrace{1 + \frac{i}{12}}_{\alpha} + \frac{i}{12} \underbrace{\left(1 + \frac{i}{12}\right)}_{\alpha} = \left(1 + \frac{i}{12}\right) \left(1 + \frac{i}{12}\right) = \underbrace{\left(1 + \frac{i}{12}\right)^2}_{\beta}$$

- after the third month, a capital equal to:

$$\underbrace{\left(1 + \frac{i}{12}\right)^2}_{\beta} + \frac{i}{12} \underbrace{\left(1 + \frac{i}{12}\right)^2}_{\beta} = \left(1 + \frac{i}{12}\right)^2 \left(1 + \frac{i}{12}\right) = \left(1 + \frac{i}{12}\right)^3$$

Repeating the argument, the capital at the end of the year therefore depends on the rate i according to the formula:

$$k(i) = \left(1 + \frac{i}{12}\right)^{12} \quad \text{with } i \in [0, 1]$$



Hence, the rate i is associated with the capital $k(i)$ at the end of the year: $i \mapsto k(i)$

- In each of the previous examples, a *real number* (**input**) is associated *uniquely* with another *real number* (**output**). It is precisely the **uniqueness of the relation** that characterizes a function.
- In general, the **admissible inputs** for a given relation (function) are subject to natural restrictions, related to the very nature of the relation.

Example 3: Admissible inputs

- In the first example the “starting” real number has the meaning of *time*; imagining that the object is dropped at an initial time $t = 0$, it is clear that we must restrict ourselves to times $t \geq 0$.
- In the second example, clearly we must have $0 \leq i \leq 1$, where 1 corresponds to a rate of 100%.

2 Definition of function, domain, codomain and image

Definition 1: of domain

The set of admissible inputs for a given function is called the **domain**.

- The expressions **independent variable** (for a generic *input*) and **dependent variable** (for the *output*) are often used.

Example 4: Other types of relations

Let us consider the set of students of a given course.

- If A is the set of students and B is the set of their names, the relation that associates each student with his or her name is *unique* (single-valued).
- In this case neither A nor B is a set of numbers, but nevertheless a unique (single-valued) relation between these two sets is well defined.

On the other hand, the converse is not necessarily true: there could be two students with the same name.

Definition 2: of function

Given any two sets A , B , a **function** f with **domain** A and values in B (or also “with **codomain** B ”) is any law that associates with each element of A one and only one element of B .

- The notation:

$$f : A \rightarrow B$$

(which reads “ f defined from A to B ”), indicates the **domain** and the **codomain** of the function f .

- The notation

$$f : x \mapsto f(x)$$

(which reads “ f maps x to $f(x)$ ”) indicates how the function f acts on the elements.

- The **symbol** $f(x)$ **denotes the output or the value** that the function f associates with x , and it must not be confused with the symbol f , which denotes the **function itself**.

Example 5: Notation for functions

For the function

$$k(i) = \left(1 + \frac{i}{12}\right)^{12}$$

of the previous example we would have:

$$k : [0, 1] \rightarrow \mathbb{R}$$

$$k : i \mapsto \left(1 + \frac{i}{12}\right)^{12}$$

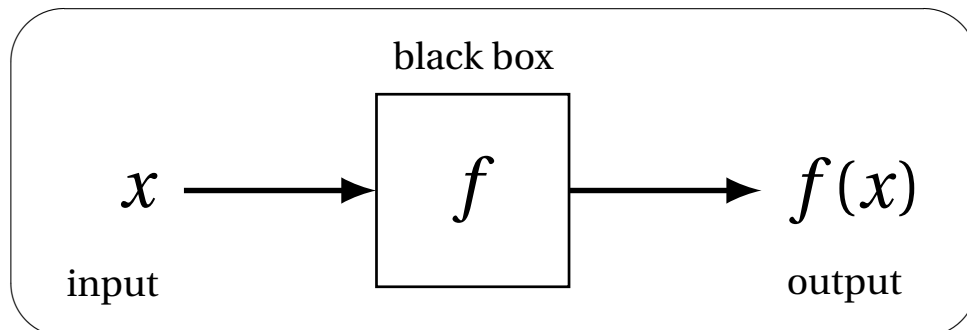
- The notation

$$y = f(x)$$

is also used to indicate that the variable y is a function of x . In this case y is the output or the value that f associates with the input x .

For brevity, sometimes the value of $f(x)$ is defined directly in terms of x , as for example $f(x) = x^2 + 10$.

- In general, we can think of a function as a **black box** that associates with each admissible input x (**input**) a unique output $f(x)$ (**output**):



Definition 3: of image and image of the domain

The output corresponding to x is called the **image** of x ; the set of possible outputs is called the **image of the domain A under f** and is denoted by the symbol $f(A)$ or $\text{Im } f$.

- Note that in the notation $f : A \rightarrow B$, the codomain B can be larger than the image $f(A)$, that is, in general we have

$$f(A) \subseteq B.$$

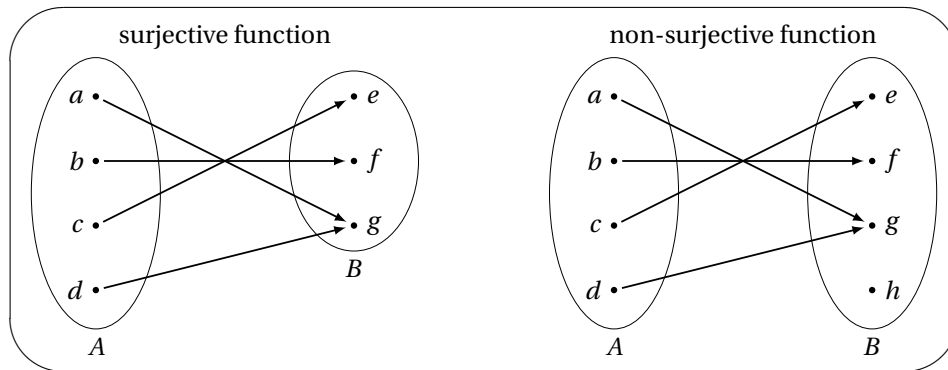
- If f is real-valued, we usually write $f : A \rightarrow \mathbb{R}$ without specifying the actual image of f .

3 Surjections, injections and bijections

Definition 4: of surjection (surjective function)

A function f is a **surjection** if the image of its domain coincides with its codomain.

- A surjection $f : A \rightarrow B$ is sometimes also called a mapping *onto* B (from A).
- Set representation:



Example 6: surjective and non-surjective functions

- The function $f(n) = \lfloor \frac{n}{2} \rfloor$ is a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since every element of the codomain $\mathbb{N}$ is the image of some value of the domain.
- The function $f(n) = 2n$ is not a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since no argument of f produces 3 as a value.
- The function $f(n) = 2n$ is, however, a surjective function from the natural numbers to the even numbers.

Remark 1

Given two sets A and B , and a function $f : A \rightarrow B$, if f is surjective then $|A| \geq |B|$.

Proof. We proceed by induction on the number n of elements in the codomain of the function.

- **Base case**

If there is only one element in the codomain ($|B| = 1$), since the function is surjective, we have $|A| \geq 1$ (every element of B is the image of at least one element of A). Hence $1 \geq 1$, which is clearly true.

- **Inductive step**

Assume then that all functions with codomain of size n satisfy $|A_n| \geq |B_n|$.

Considering codomains of size $n + 1$ we have $|B_{n+1}| = |B_n| + 1$. Since the codomain has one more element and the function is surjective, we have $|A_{n+1}| \geq |A_n| + 1$. Substituting we obtain:

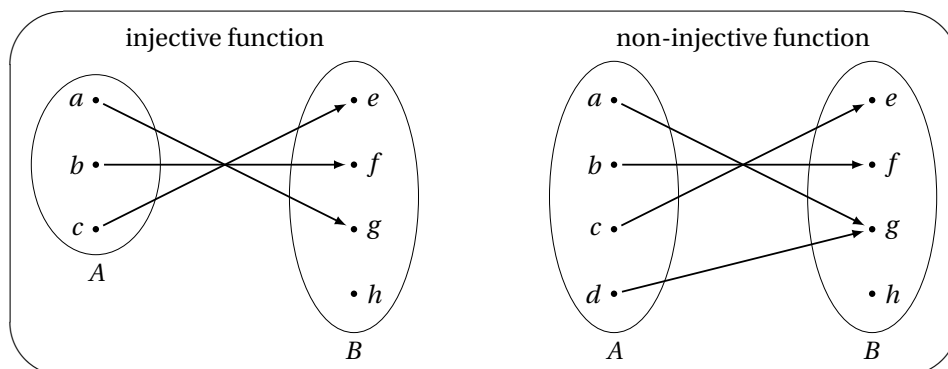
$$\underbrace{|A_n|}_{\leq |A_{n+1}| - 1} \geq \underbrace{|B_n|}_{= |B_{n+1}| - 1} \quad \text{hence } |A_{n+1}| \geq |B_{n+1}|.$$

□

Definition 5: of injection (injective function)

A function f is an **injection** if distinct arguments of f produce distinct values, that is, if $a \neq b$ implies $f(a) \neq f(b)$.

- set representation:



Example 7:

- The function $f(n) = 2n$ is an injective function from $\mathbb{N}$ to $\mathbb{N}$, since each even number b is the image under f of exactly one element of the domain, namely $n = \frac{b}{2}$
- The function $f(n) = \lfloor \frac{n}{2} \rfloor$ is not an injective function, since the value 1 is obtained from two arguments: 2 and 3.

- An injection is also called a **one-to-one** function.

Remark 2

Given two sets A and B , and a function $f : A \rightarrow B$, if f is injective then $|A| \leq |B|$.

Proof. We proceed by induction on the number n of elements in the domain of the function.

- **Base case**

If there is only one element in the domain ($|A| = 1$), since each element of the domain is associated with one and only one element of the codomain, we have $|B| \geq 1$. Hence $1 \leq 1$, which is clearly true.

- **Inductive step**

Assume then that all functions with domain of size n satisfy $|A_n| \leq |B_n|$.

Considering domains of size $n + 1$ we have $|A_{n+1}| = |A_n| + 1$ and, since the function is injective, $|B_{n+1}| \geq |B_n| + 1$. Substituting we obtain

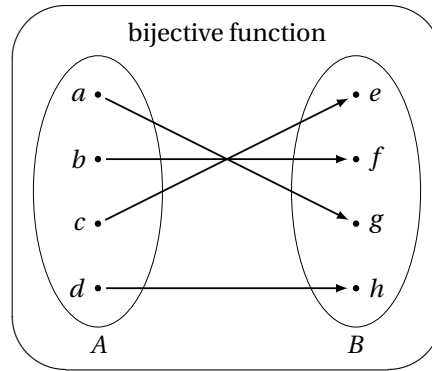
$$\underbrace{|A_n|}_{= |A_{n+1}| - 1} \leq \underbrace{|B_n|}_{\leq |B_{n+1}| - 1} \quad \text{hence} \quad |A_{n+1}| \leq |B_{n+1}|.$$

□

Definition 6: of bijection (bijective function or one-to-one correspondence)

A function f is a **bijection** if: (i) it is injective and (ii) it is surjective.

- set representation:



Example 8:

- The function $f(n) = (-1)^n \lceil \frac{n}{2} \rceil$ is a bijection from $\mathbb{N}$ to $\mathbb{Z}$. The values of the function are:

$$f(0) = 0, \quad f(1) = -1, \quad f(2) = 1, \quad f(3) = -2, \quad f(4) = 2 \dots$$

The function is injective, since no element of $\mathbb{Z}$ is the image of more than one element of $\mathbb{N}$. The function is surjective, since all elements of $\mathbb{Z}$ are images of some element of $\mathbb{N}$.

- A bijection is also called a **one-to-one** correspondence, since it pairs elements of the domain with elements of the codomain.
- A bijection from a set A to itself is also called a **permutation**.